MATH5301 Lecture Notes

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Contents

1 Lecture 9 - 22/09/26 **2**

2 Lecture 10 - 24/09/26 **2**
2.1 Compact Sets 2

3 Lecture 11 - 29/09/26 **3**

4 Lecture 12 - 01/10/26 **4**

1 Lecture 9 - 22/09/26

Definition 1.1. $A \subseteq S$ is totally bounded if

$$\forall_{\epsilon>0}\exists_{a_1,\ldots,a_n\subset A} A \subset \bigcup B_\epsilon(a_i)$$

Definition 1.2. A is sequentially compact if

$$\forall_{\{x_n\}\subset A}\exists_{\{n_k\}\subset\mathbb{N}} \lim_{k\to\infty} x_{n_k} = x_0 \in A$$

Theorem 1.3. *(S,d) metric space, $A \subset S$. A is complete and totally bounded $\implies$ A is sequentially compact.*

Proof. We will build the subsequence. Pick $\{x_n\} \subset A$.
First, we prove that if A is totally bounded $\implies$ $\forall_{k\in\mathbb{N}}\exists_{a_k\in A}$ ($B_{\frac{1}{k}}(a_k)$ contains an infinite number of elements of $\{x_k\}$). Assume there does not exist an open ball with an infinite number of elements. Since A has a finite cover, there are a finite number of open balls covering A. But then, none of the balls can contain an infinite number of elements by assumption, contradiction.

1. Fix $k = 1$. By the above, we have some $B_1(a_1)$ with infinite elements of $\{x_n\}$. Choose some $x_1 \in \{x_n\}$. Delete all elements of $\{x_n\}$ outside $B_1(a_1)$ but keep x_1. Call the new sequence $\{x_n\}$ again.
$$d(x_n, x_m) < d(x_n, a_1) + d(x_m, a_1) < 1 + 1 = 2, \quad n, m > 1$$
2. Now consider $k = 2$. Apply the above lemma to the new sequence, which implies $B_{\frac{1}{2}}(a_2)$ contains an infinite number of elements from the new sequence. Now, choose $x_2 \in \{x_n\}$, which is the sequence. Delete all elements of $\{x_n\}$ outside the new ball, but keep x_1 and x_2. Call the new sequence $\{x_n\}$ again.
$$d(x_n, x_m) < \frac{1}{2} + \frac{1}{2} = 1, \quad n, m > 2$$
n. $k = N$. Apply the lemma, delete all elements outside $B_{\frac{1}{N}}(a_n)$ but keep $x_1, \ldots, x_N$.
$$d(x_n, x_m) < 2 \cdot \frac{1}{N}, \quad n, m > N$$
We save the elements x_n each time to make sure our building sequence still has countably many elements left even after deleting.

After the N steps, we have a Cauchy subsequence of the original sequence. Since complete space, it is convergent to x_0. Thus, sequentially compact satisfied. □

2 Lecture 10 - 24/09/26

2.1 Compact Sets

Definition 2.1 (7.4)**.** Let $\{U_\lambda\}_{\lambda\in\Lambda}$ be a family of subsets of a metric space (S,d).

1. $\{U_\lambda\}$ is a cover of A if $A \subset \bigcup A_\lambda$

2. $\{U_\lambda\}$ is an open cover if $\{U_\lambda\}$ is a cover and each U_λ is open.
3. $\{U_\lambda\}_{\lambda\in\Lambda'}$ is a subcover of $\{U_\lambda\}$ if $\Lambda' \subset \Lambda$.
4. $\{U_\lambda\}$ is a finite cover if Λ is finite.

Definition 2.2 (7.5)**.** A is compact if every open cover of A contains a finite subcover.

Example 2.3 (7.1)**.** If A and C are compact, then $A \cup C$ is compact.

Proof. Fix an arbitrary open cover $\{U_\lambda\}_{\lambda\in\Lambda}$ of $A \cup C$. Then this cover is an open cover of A. Since A is compact, there exists a finite $\Lambda' \subset \Lambda$ such that $\{U_\lambda\}_{\lambda\in\Lambda'}$ is a cover of A. Do the same for C. Then the union of the two finite subcovers covers $A \cup C$. □

Lemma 2.4 (Lebesque covering lemma)**.** *Let $\{U_\lambda\}$ be an open cover of A. If A is sequentially compact, then*

$$\exists_{\epsilon>0}\forall_{x\in A}\exists_{\lambda\in\Lambda} B_\epsilon(x) \subset U_\lambda$$

Proof. Assume the contrary

$$\forall_{\epsilon>0}\exists_{x\in A}\forall_{\lambda\in\Lambda} B_\epsilon(x) \not\subset U_\lambda$$

$$\forall_{\frac{1}{n}>0}\exists_{x_n\in A}\forall_{\lambda\in\Lambda} B_{\frac{1}{n}}(x_n) \not\subset U_\lambda$$

By definition of sequentially compact,

$$\{x_n\} \subset A \implies \exists_{\text{convergent } \{x_{n_k}\}\subset\{x_n\}} \wedge \lim_{k\to\infty} x_{n_k} = x_n \in A$$

$$x_0 \in A \subset \bigcup U_\lambda \implies x_0 \in \bigcup U_\lambda \implies \exists_{\lambda\in\Lambda} x_0 \in U_{\lambda_0}$$

We claim $B_{\frac{1}{n_k}}(x_{n_k}) \subset U_{\lambda_0}$ for sufficiently large $k \in \mathbb{N}$. We want to estimate the distance between the ball and x_0. Pick $y \in B_{\frac{1}{n_k}}(x_{n_k})$ and compute $(y, x_0) \le d(y, x_{n_k}) + d(x_{n_k}, x_0)$ note that

$$d(y, x_{n_k}) < \frac{1}{n_k}, \quad d(x_{n_k}, x_0) \to 0 \text{ as } k \to \infty$$

Since U_{λ_0} is open,

$$\exists_{r>0} B_r(x_0) \subset U_{\lambda_0}$$

Now we have to observe that for sufficiently large k,

$$d(y, x_{n_k}) < \frac{r}{2} \wedge d(x_{n_k}, x_0) < \frac{r}{2}$$

□

3 Lecture 11 - 29/09/26

Theorem 3.1 (7.1)**.** *If A is complete and totally bounded, A is sequentially compact.*

Theorem 3.2 (8.1)**.** *If A is complete and totally bounded, A is compact.*

Proof. Let $\{U_\lambda\}$ be an arbitrary open cover of A. Need to prove $\exists_{\text{ finite } \Lambda' \in \Lambda} \bigcup_{\lambda in \Lambda'} U_\lambda \supset A$.
By Theorem 7.1, A is sequentially compact.
Then by Lebesque covering lemma, $\exists \epsilon > 0, \forall x \in A \exists_{\lambda \in \Lambda} B_\epsilon(x) \subset U_\lambda$.
By total boundedness of A, $\exists_{\{a_1,\ldots,a_n\} \subset A} \bigcup_{i=1}^n B_\epsilon(a_i) \supset A$.
Then $\forall_{i \in 1,n} \exists_{\lambda_i \in \Lambda} \exists_{\lambda_i \in \Lambda} B_\epsilon(a_i) \subset U_{\lambda_i}$.
Then $\Lambda' = \{\lambda_1, \ldots, \lambda_n\}$ is the required one for compactness. □

Theorem 3.3 (9.2)**.** *If A is sequentially compact, then A is totally bounded and complete.*

Proof. Completeness. Take an arbitrary Cauchy sequence $\{x_n\} \subset A$.
From sequential compactness, $\exists_{\{x_{n_k}\} \subset \{x_n\}} \lim_{k \to \infty} x_{n_k} = x_0 \in A$.
Therefore the limit of $\{x_n\}$ exists and $\lim_{n \to \infty} \{x_n\} = x_0$.
Total boundedness. Assume A is not totally bounded. Then $\exists_{\epsilon > 0} \forall_{\{a_1,\ldots,a_n\} \subset A} A \backslash \bigcup_{i=1}^n B_\epsilon(a_i) \neq \emptyset (*)$
Fix $x_1 \in A$.

Step 1: Apply $(*)$ with $\{a_1, \ldots, a_n\} = \{x_1\}$. By the statement, $A \backslash B_\epsilon(x_1) \neq \emptyset$. Then pick from non-empty set $x_2 \in A \backslash B_\epsilon(x_1)$.

Step 2: Apply $(*)$ with $\{a_1, \ldots, a_n\} = \{x_1, x_2\}$. By (*), $A \backslash B_\epsilon(x_1) \cup B_\epsilon(x_2) \neq \emptyset$. Pick $x_3 \in A \backslash B_\epsilon(x_1) \cup B_\epsilon(x_2)$.

continue building sets then $x_{k+1} \in A \backslash \bigcup^k B_\epsilon(x_i)$. Then distance $d(x_{k+1}, x_i) > \epsilon$. We claim none of the subsequences from our constructed sequence are convergent. Then the distance shows that for any subsequence, it is not Cauchy and does not converge to element in S This contradicts sequential compactness. □

Theorem 3.4 (9.1)**.** *If A is compact, A is sequentially compact.*

Proof. Assume the contrary, $\exists_{\{x_n\}_{n=1}^\infty \subset A}$ s.t. $\{x_n\}$ doesn't contain a convergent subsequence. $(*)$
We claim $(*)$ implies elements of $\{x_n\}$ are isolated, i.e.

$$\forall_{n \in \mathbb{N}} B_\epsilon(x_n) \cap \{x_n\}_{n=1}^\infty = \{x_n\} \text{ where } \{x_i\}_{n=1}^\infty = \bigcup \{x_i\}$$

Assume by contradiction that this doesnt hold. Then

$$\exists_{n_* \in \mathbb{N}} \forall_{\epsilon > 0} B_\epsilon(x_{n_k}) \cap \{x_n\}_{n=1}^\infty \neq \{x_{n_*}\}$$

Consider the following covering of A: $\{U_i\}_{i \in \overline{0,\infty}}$ where $U_i = B_{\epsilon_i}(x_i), i \in \overline{1,\infty}$, and $U_0 = S \backslash \{x_n\}_{n=1}^\infty$.
Note that U_0 is open if $\{x_n\}_{n=1}^\infty$ is closed. Recall that if a set has a limit point outside of itself, the set is not closed. Since $\{x_n\}_{n=1}^\infty$ may have limit points not in A, U_0 is not guarenteed to be open. Thus, if we modify U_0 to have removed the limit points,

$$U_0 = S \backslash \{x_n\}_{n=1}^\infty \cup (\{x_n\}_{n=1}^\infty)'$$

We notice that all of the limit points are outside of A. □

4 Lecture 12 - 01/10/26